Indefinite Integration: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Meaning of integration

Formulas

$$\frac{d}{dx}\phi(x) = f(x) \iff \int f(x)\,dx = \phi(x) + C$$

KEY POINTS

- f = integrand, ϕ = primitive / anti-derivative, C = constant of integration
- Two primitives of f differ by a constant; a given value $f(a) = k$ fixes C
- Check any answer by differentiating it

2. Standard results

Formulas

$\int x^n dx = \dfrac{x^{n+1}}{n+1}$ $(n \neq -1)$, $\int \dfrac{dx}{x} = \ln|x|$, $\int a^x dx = \dfrac{a^x}{\ln a}$

$\int \dfrac{dx}{a^2+x^2} = \dfrac{1}{a}\tan^{-1}\dfrac{x}{a}$, $\int \dfrac{dx}{x^2-a^2} = \dfrac{1}{2a}\ln\left|\dfrac{x-a}{x+a}\right|$

$\int \tan x = \ln|\sec x|$, $\int \cot x = \ln|\sin x|$, $\int \sec x = \ln|\sec x + \tan x|$, $\int \operatorname{cosec} x = \ln|\tan\frac{x}{2}|$

KEY POINTS

- $ax+b$ in place of x: divide the result by a
- Simplify first: $e^{\ln f} = f$, identities, product to sum, $\sin^2, \cos^2, \sin^3, \cos^3$ to multiple angles
- linear/linear = constant $+\dfrac{k}{\text{linear}}$; $\dfrac{1}{1 \pm \sin x}$: multiply by conjugate

3. Substitution

Formulas

$$\int f(\phi(x))\phi'(x)\,dx = \int f(t)\,dt, \quad \phi(x) = t$$

$\int \dfrac{f'}{f} = \ln|f|$, $\int f^n f' = \dfrac{f^{n+1}}{n+1}$

KEY POINTS

- Look for a factor that is the derivative of an inside expression
- $\frac{1}{x}$ with $\ln x$: put $\ln x = t$; $\frac{1}{1+x^2}$ with $\tan^{-1} x$: put $\tan^{-1} x = t$
- $\dfrac{1}{a\sin x + b\cos x}$: write $r\sin(x+\alpha)$, integrate cosec
- Exponentials: put $e^x = t$ (or divide by the largest power of e^x)

4. Trigonometric substitution and square roots

Formulas

$a^2+x^2 : x = a\tan\theta$; $a^2-x^2 : x = a\sin\theta$; $x^2-a^2 : x = a\sec\theta$

$\int \dfrac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\dfrac{x}{a}$, $\int \dfrac{dx}{\sqrt{x^2 \pm a^2}} = \ln\left|x + \sqrt{x^2 \pm a^2}\right|$

KEY POINTS

- $\sqrt{\frac{a-x}{a+x}}$: $x = a\cos 2\theta$; $\sqrt{\frac{x}{a-x}}$: $x = a\sin^2\theta$
- $\dfrac{1}{\text{Quad}}$, $\dfrac{1}{\sqrt{\text{Quad}}}$, $\sqrt{\text{Quad}}$: complete the square first

5. Integration by parts

Formulas

$$\int \mathrm{I}\cdot\mathrm{II}\,dx = \mathrm{I}\int \mathrm{II} - \int\left(\mathrm{I}'\int \mathrm{II}\right)dx$$

$$\int e^{ax}\sin bx = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2},\ \int e^{ax}\cos bx = \frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2}$$

$$\int \sqrt{a^2-x^2} = \tfrac{x}{2}\sqrt{a^2-x^2} + \tfrac{a^2}{2}\sin^{-1}\tfrac{x}{a},\ \int \sqrt{x^2\pm a^2} = \tfrac{x}{2}\sqrt{x^2\pm a^2} \pm \tfrac{a^2}{2}\ln\left|x+\sqrt{x^2\pm a^2}\right|$$

KEY POINTS

- First function by ILATE: Inverse, Log, Algebraic, Trig, Exponential
- Single function ($\ln x$, $\tan^{-1}x$): take 1 as second function
- Cyclic: integral comes back, move it to the left; I_n relations come from parts

6. Two classic integrals

Formulas

$$\int e^x[f+f']dx = e^x f + C,\ \int[f+xf']dx = xf + C$$
$$\int e^{kx}[kf+f']dx = e^{kx}f,\ \int e^g(fg'+f')dx = e^g f$$

KEY POINTS

- Split the bracket into a function plus its derivative (break fractions, use identities)
- Exponent other than x: put the exponent $= t$

7. Rational functions

KEY POINTS

- 1/linear: $\frac{1}{a}\ln|ax+b|$; poly/linear or poly/quad: divide first
- linear/Q, linear/$\sqrt{Q}$, linear$\cdot\sqrt{Q}$: linear $= A\,Q' + B$
- $\int \frac{Q'}{\sqrt{Q}} = 2\sqrt{Q}$, $\int Q'\sqrt{Q} = \frac{2}{3}Q^{3/2}$

8. Partial fractions

Formulas

$$\frac{P}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b},\ \frac{P}{(x-a)^2(x-b)} = \frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \frac{B}{x-b}$$

$$\frac{P}{(x-a)(x^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$$

STEPS

1. Improper ($\deg P \geq \deg Q$)? Divide first
2. Substitute ($x^2 = t$, $e^x = t$, $\sin x = t$) to make it rational if needed
3. Cover-up: put $x = a$ for $\frac{A}{x-a}$; compare coefficients for the rest

9. Trigonometric integrands

KEY POINTS

- $\sin^m x\cos^n x$: m odd $\to \cos x = t$; n odd $\to \sin x = t$; both even $\to$ multiple angles; $m+n$ negative even $\to \tan x = t$
- $\frac{1}{a\sin^2 x + b\cos^2 x + c\sin x\cos x + d}$: multiply by $\sec^2 x$, $\tan x = t$
- $\frac{1}{a\sin x + b\cos x + c}$: $\tan\frac{x}{2} = t$
- $\frac{l\sin x + m\cos x + n}{a\sin x + b\cos x + c}$: Nr $= A\,\mathrm{Dr} + B\,\mathrm{Dr}' + C$
- $(\sin x + \cos x)$ over $\sin 2x$: $\sin x - \cos x = t$; $(\sin x - \cos x)$ over $\sin 2x$: $\sin x + \cos x = t$
- $\sin 2x$ over $(\sin x \pm \cos x)$: $\sin 2x = (\sin x + \cos x)^2 - 1 = 1 - (\sin x - \cos x)^2$

10. Miscellaneous profiles

KEY POINTS

- Kutur putur: $x(x^n+1)$, $x^2(x^n+1)^{(n-1)/n}$, $x^n(1+x^n)^{1/n}$ in Dr: take x^n common, $1+x^{-n}=t$
- $\dfrac{x^2 \pm k}{x^4+px^2+k^2}$: divide by x^2; $x^2+k \to x-\frac{k}{x}=t$, $x^2-k \to x+\frac{k}{x}=t$
- $\sqrt{\tan x}$, $\dfrac{1}{1+x^4}$, $\dfrac{x^2}{1+x^4}$ reduce to this profile
- $\dfrac{1}{(x-a)^m(x-b)^n}$, $m+n=2$: $\dfrac{x-a}{x-b}=t$

11. Irrational algebraic functions

KEY POINTS

- $(x-\alpha)(\beta-x)$: $x=\alpha\cos^2\theta+\beta\sin^2\theta$
- $(x-\alpha)(x-\beta)$: $x=\alpha\sec^2\theta-\beta\tan^2\theta$ (or $x-\alpha=t^2$)
- $\dfrac{1}{(\text{lin or quad})\sqrt{px+q}}$: $px+q=t^2$
- $\dfrac{1}{(ax+b)\sqrt{px^2+qx+r}}$: $ax+b=\frac{1}{t}$
- $\dfrac{1}{(ax^2+b)\sqrt{px^2+q}}$: $x=\frac{1}{t}$, then $p+qt^2=y^2$

12. Common mistakes checklist

STEPS

1. Never forget $+C$; use the given value to find C
2. $\int \dfrac{dx}{x}=\ln|x|$ (modulus), $\sqrt{x^2}=|x|$
3. Divide by the coefficient of x for $f(ax+b)$
4. $\dfrac{x^{n+1}}{n+1}$ fails for $n=-1$
5. Answers in different forms can differ by a constant; compare by differentiating
6. Change dx fully when substituting; no x may remain